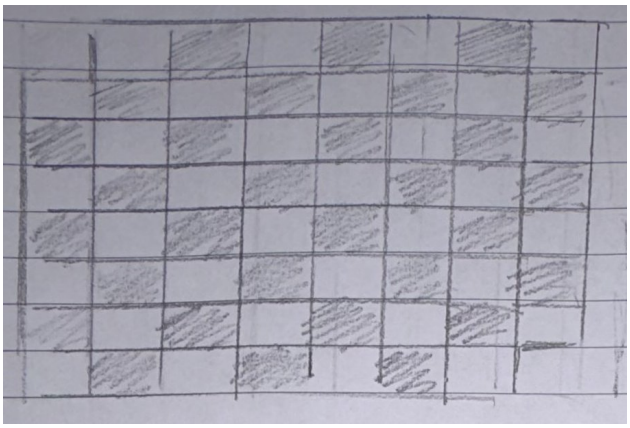
WMC Junior Finals Proof Questions - Set 2

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1 A Lesson in Invariants

Problem 1. There is an 8×8 chessboard with two tiles missing, as indicated below.



Prove that it is impossible to tile the board with 2×1 tiles.

Solution 1.

(a) Note that any 2×1 tile will cover 1 white and 1 black square. Hence, any tiling of a chessboard with 2×1 tiles will cover an equal number of white and black squares.

A normal chessboard has

$$\frac{1}{2}(8)(8) = 32$$

white squares and 32 black squares. The modified board has 32 white squares, but only

$$32 - 2 = 30$$

black squares: not equal.

Hence, it is impossible to tile the board with 2×1 tiles. $\square$

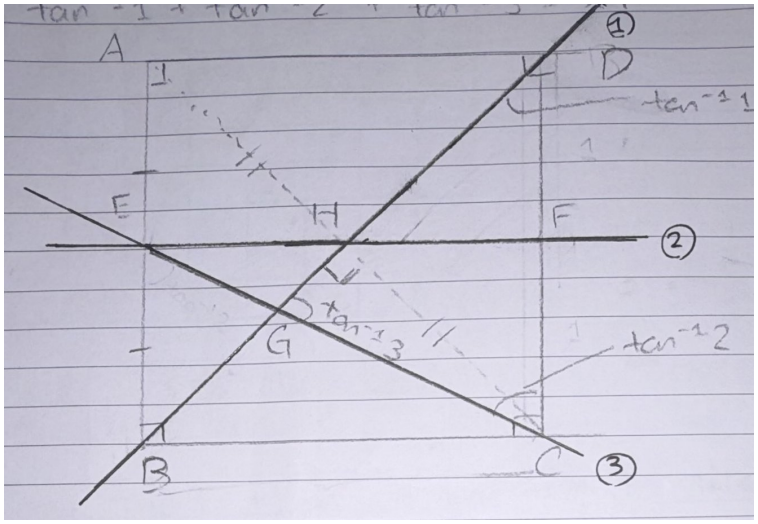
2 Trigonometry with Paper Folding

Problem 2. By folding a square piece of paper, prove that

$$\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi.$$

(**Hints:** Centroids split a median into the ratio $2 : 1$ (two-thirds and one-third). Can you use that to get $\tan^{-1} 3$? What kind of special right triangles have $\tan^{-1} 1$ and $\tan^{-1} 2$ as angles? Think about the ratio of sides.)

Solution 2.



Notice that in $\triangle ABC$, $\overline{CE}$ and $\overline{BH}$ are the medians. Hence, their intersection, G, is the centroid, which divides $\overline{BH}$:

$$BG : GH = 2 : 1,$$

or $BH = 3 \cdot GH$. Hence, as $CH = BH$,

$$\tan \angle CGH = \frac{CH}{GH} = \frac{BH}{GH} = \frac{3 \cdot GH}{GH} = 3.$$

Hence,

$$\angle CGH = \tan^{-1} 3.$$

Observe that

$$\angle BDC = \tan^{-1}\left(\frac{BC}{DC}\right) = \tan^{-1} 1$$

and

$$\angle ECF = \tan^{-1}\left(\frac{EF}{CF}\right) = \tan^{-1} 2.$$

As $\triangle GCD$ is a triangle,

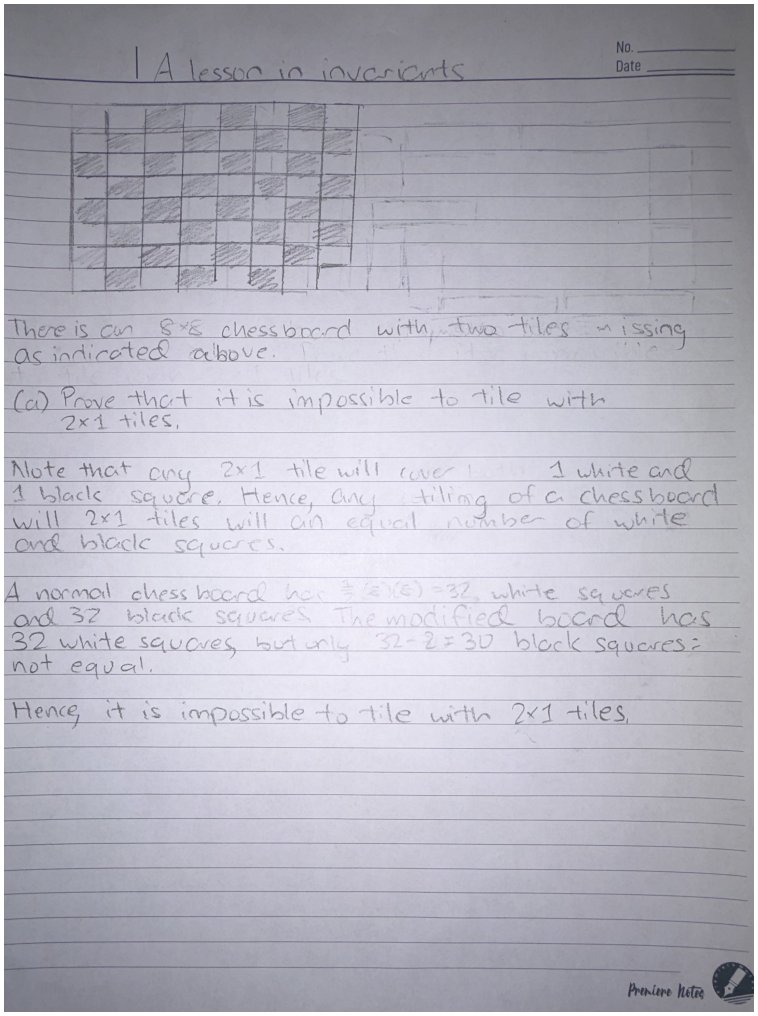
$$\angle GDC + \angle DCG + \angle CGD = \tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi. \qquad \square$$

Acknowledgements / Transparency

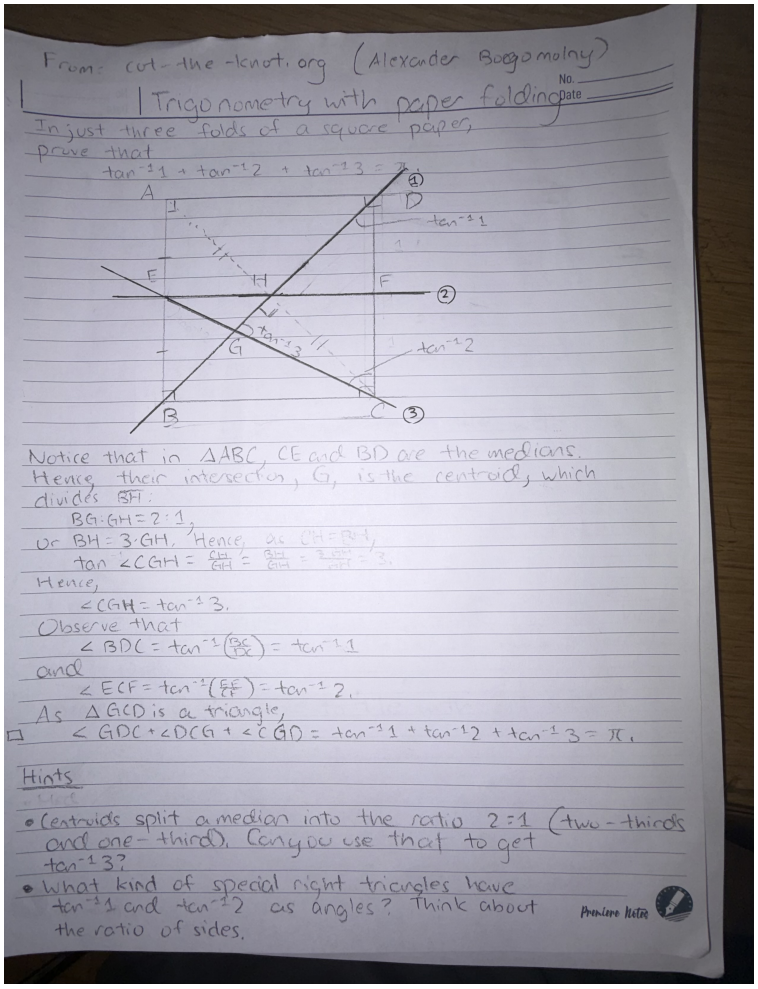
- I wrote Problem 1 from memory, but it is not the most original problem – it is a stock example in problem books which I do not think I took the time to develop in an unusually novel or clear manner.
- Problem 2 is from *cut-the-knot.org* (Alexander Bogomolny).
- I used AI to help me typeset in order to save time.

A Original Handwritten Working

A.1 Problem 1



A.2 Problem 2


From: cut-the-knot.org (Alexander Bogomolny)
Trigonometry with paper folding
No.
Date
In just three folds of a square paper, prove that
$\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$.
A
D
E
H
F
G
B
C
1
2
3
$\tan^{-1}1$
$\tan^{-1}2$
$\tan^{-1}3$
Notice that in ΔABC, CE and BD are the medians. Hence their intersection, G, is the centroid, which divides BH:
$BG:GH = 2:1$,
or $BH = 3 \cdot GH$. Hence, as $CH = BH$,
$\tan \angle CGH = \frac{CH}{GH} = \frac{BH}{GH} = \frac{3 \cdot GH}{GH} = 3$.
Hence,
$\angle CGH = \tan^{-1}3$.
Observe that
$\angle BDC = \tan^{-1}\left(\frac{BC}{DC}\right) = \tan^{-1}1$
and
$\angle ECF = \tan^{-1}\left(\frac{EF}{CF}\right) = \tan^{-1}2$.
As ΔGCD is a triangle,
$\angle GDC + \angle DCG + \angle CGD = \tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$.
□
Hints
• Centroids split a median into the ratio 2:1 (two-thirds and one-third). Can you use that to get $\tan^{-1}3$?
• What kind of special right triangles have $\tan^{-1}1$ and $\tan^{-1}2$ as angles? Think about the ratio of sides.
Premiere Notes